Li-Kang (Tony) Weng

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WORK EXPERIENCE

Software Development Engineer, Amazon Web Services, Inc. — Seattle, USA

- Tech lead of an EC2 demand-shaping team; led a cross-team initiative across 20+ partner teams that unlocked an additional **$1.8B** in CapEx savings on top of $2B+ already delivered by the broader program. — 06/2021 - present
- Led a cross-team effort to design and deliver a public instant capacity block feature on a tight re:Invent timeline, enabling **on-demand** GPU capacity access and reducing idle capacity.
- Established operational audit mechanisms and automated manual workflows using AI tooling, yielding **$3M** in annual OpEx savings and reclaiming **1 month** of engineering time per year.
- Mentored **10+** engineers across organizations, with several receiving promotions; consistently recognized by mentees for hands-on technical guidance and career growth support.

Software Development Engineer Intern, Amazon Web Services, Inc. — Dallas, USA

- Added CloudWatch Events to AWS EC2/Spot Fleet to let customers be aware of the states of their fleets and take action to fulfill their business demands — 05/2020 - 07/2020

Research Assistant, National Taiwan University — Taipei, Taiwan

- Built a tracked robot using ROS (Robot Operating System) for surveillance on a chicken farm and the information and the manipulation of the robot were visualized on the GUI — 03/2019 - 06/2019

Software Engineer, HTC Corp. — Taipei, Taiwan

- Developed real-time **visual-inertial SLAM** algorithm using C++ on ARM platform — 01/2017 - 07/2018
- Designed GUI and tools for real-time data visualization and KPI measurement to eliminate tedious labor and quantitate tracking performance

TECHNICAL SKILLS

Java, Python, TypeScript, Ruby, JavaScript, C++, AWS, Git, Prompt Engineering

EDUCATION

The University of Texas at Dallas — Dallas, USA
Master of Science in Computer Science — 05/2021

National Taiwan University (NTU) — Taipei, Taiwan
Master of Science in Bio-Industrial Mechatronics Engineering (BIME) — 09/2015

- **Dissertation** Sensor Fusion of Stereo Vision and Radar Systems for Vehicle Safety Application

Bachelor of Science in BIME — 06/2013

- **Topic** Quantitative Evaluation of the Floral Shape Variation in *Sinningia Speciosa* Domestication

RELEVANT PROJECTS AND RESEARCH

Sensor Fusion Project, Biophotonics and Bioimaging Laboratory (BBLab), NTU — 05/2013 - 09/2015

- Constructed **sensor fusion based** vehicle safety real-time system capable of obstacle detection, tracking and collision avoidance algorithms using **stereo vision** and **millimeter-wave radar sensor**
- Eliminated measurement error of depth information from 2.4% to **0.7%** using fused information
- Enhanced obstacle matched rate from 82.1% to **89.8%** using fused information
- Accelerated **2.8** times in the correspondence matching method using CUDA with OpenCV

The 9th Utechzone Machine Vision Prize, Utechzone Inc. — 02/2014 - 08/2014

- Awarded **2nd prize** in the overall competition and developed fall detection algorithm including background removal, feature extraction, object tracking, and motion detection under complicated scenarios (light variation, overlap)

Floral Shape Variation Study, BBLab, NTU — 08/2012 - 02/2014

- **Accelerated process speed** and **eliminated measurement error** by developing a semi-automatic program with GUI using image processing methods for flower landmark acquisition
- Analyzed shape variation of *Sinningia speciosa* from landmarks identified on 2D images

Advanced Technology Project in Vehicle Safety: Intelligence and Human Factors, ARTC — 05/2013 - 01/2014

- Refactored obstacle matching algorithm to optimize performance by speed-up around **30%**
- Eliminated measurement error by optimizing camera calibration on stereo vision
- Analyzed **path planning** algorithms to provide a more realistic solution
- Designed GUI with concise information for users easy to understand environmental information

SELECTED PUBLICATIONS

Ta-Te Lin, **Li-Kang Weng** and An-Chih Tsai. 2014. Object Tracking and Collision Avoidance Using Particle Filter and Vector Field Histogram Methods. Paper presented at ASABE. Paper No. 1906189, Montreal, Quebec City, Canada.